Carlink Incident Reporting Dashboard — Product Spec

1. Overview

Build a web dashboard for incident reporting that stores structured incident data, photo evidence, AI-generated analysis, human edits, and the final PDF report.

This system should support the current **security incident** workflow and be designed so it can later expand into **vehicle damage / claim reporting**. The dashboard is the working system for review, editing, tracking, and analytics. The PDF is the official final output that can be downloaded, printed, or shared. The current report template includes reporter info, incident details, description, witnesses, incident category, immediate actions, reporting to authorities, preventive measures, and additional comments.

2. Product Goals

1. Capture incident reports in a structured way.
2. Store photos and evidence linked to each incident.
3. Use AI to draft and classify incident details.
4. Allow human review and correction before sign-off.
5. Generate a professional PDF report from the final data.
6. Provide dashboard analytics and searchable history.

The broader system proposal emphasizes that AI should draft reports, but a human must still sign off before finalization. It also describes the dashboard as the place for list, search, export, and review workflows.

3. User Roles

Admin

- Full access to all incidents
- Can edit, review, sign off, and export
- Can manage users and settings

Surveyor / Reviewer

- Reviews AI drafts
- Edits incident details
- Approves or rejects reports
- Signs off final report

Viewer

- Read-only access
- Can search and open reports
- Can download PDF if permitted

4. Core Workflow

1. User submits incident photos and description.
2. AI extracts structured data from text and photos.
3. System creates a draft incident record.
4. Reviewer checks and edits the draft.
5. Reviewer confirms the incident.
6. System generates the official PDF.
7. Report is stored and made available in the dashboard.

5. Main Dashboard Pages

5.1 Dashboard Home

Show a high-level summary:

- Total incidents
- Open / pending review
- Signed-off reports
- High severity incidents
- Incidents this month
- Average resolution time
- Category breakdown
- Recent activity

5.2 Incident List

A searchable, filterable list of all incidents.

Columns:

- Incident ID
- Date
- Time
- Category
- Severity
- Status
- Reporter
- Location
- Assigned reviewer
- Last updated

Filters:

- Date range
- Category
- Severity
- Status
- Reporter
- Location
- AI confidence
- Has photos / no photos

5.3 Incident Detail Page

This is the most important page.

It should show:

- Full incident summary
- All extracted structured fields
- AI suggestions
- Human edits
- Photo gallery
- Witnesses
- Actions taken
- Authority report status
- Preventive measures
- PDF preview
- Audit trail

5.4 Analytics Page

Show trends:

- Incidents over time
- Incident type distribution
- Severity distribution
- Location breakdown
- Resolution time
- Most common damage / incident patterns
- AI correction frequency

5.5 Settings / Admin Page

- User management
- Role permissions
- Report template settings
- AI prompt / model settings
- PDF branding settings

• Retention and export settings

6. Incident Data Fields

The system should store the following data.

6.1 Reporter Information

• Name
• Job title / role
• Contact details
• Channel source
• Reporter ID

The uploaded template includes report filer name, job title / role, and contact details.

6.2 Incident Details

• Incident date
• Incident time
• Location
• Category
• Subcategory
• Severity
• Description
• Free-text notes

The template explicitly includes date/time, location, incident description, and category selection.

6.3 People Involved

• Reporter
• Victim / affected person
• Suspected person
• Witnesses
• Contact details
• Statements

6.4 Actions Taken

• Immediate actions
• Area secured
• Security informed
• Medical assistance provided
• Police / authority report status
• Case / reference number

The template asks for immediate actions taken and whether the incident has been reported to authorities.

6.5 Preventive Measures

- Recommended actions
- Status of each recommendation
- Responsible person
- Due date

The template includes a preventive measures section.

6.6 Evidence / Photos

- Photo file
- Upload time
- Uploaded by
- AI caption
- AI tags
- Linked damage / incident item
- Human verification status

7. Category Design

The dashboard should support a flexible category system.

Primary Categories

- Unauthorized Access
- Theft / Burglary
- Vandalism / Property Damage
- Assault / Threat
- Harassment
- Cybersecurity Breach
- Workplace Accident
- Vehicle Damage / Traffic Incident
- Fire / Safety Incident
- Environmental Incident
- Other

Secondary Tags

Examples:

- Door forced open
- Missing equipment
- Broken glass
- Rear impact
- Front bumper damage
- Laptop theft
- CCTV issue

- Unauthorized entry

The category system must allow:

- Single category or multi-category selection
- Custom “Other” text
- Later expansion for vehicle damage workflows

8. Severity Design

Use a 3-level severity scale:

- Minor
- Moderate
- Severe

Optionally add:

- Critical

Severity should be:

- AI-suggested
- Human-editable
- Visible on the list page and detail page

9. Photo / AI Damage Tagging

The AI should extract useful visual tags from each uploaded image.

For each photo, store:

- Caption
- Tags
- Confidence score
- Suggested linked incident item
- Suggested damage label
- Whether human confirmed it

For vehicle-related incidents later, support part-level damage tagging such as:

- Front bumper
- Rear bumper
- Left door
- Right door
- Headlight
- Windshield
- Wheel

- Side mirror

The proposal explicitly states the long-term goal is damage-item recognition and structured tagging of damaged items, with human review still required.

10. PDF Output

The PDF is not the working interface. It is the final official document.

The PDF should include:

- Report title
- Incident ID
- Reporter info
- Incident details
- Category
- Description
- Witnesses
- Immediate actions taken
- Authority report status
- Preventive measures
- Additional comments
- Photo attachments
- Reviewer / sign-off section

The PDF should be:

- Professional
- Printable
- Shareable
- Versioned
- Locked after sign-off unless reopened by admin

11. Audit Trail

Every important action must be logged:

- Incident created
- AI draft generated
- Field edited
- Photo added
- Reviewer approved
- PDF generated
- PDF re-generated
- Report signed off

Each log entry should include:

- Timestamp
- Actor
- Action
- Field changed
- Old value
- New value

12. Required Functional Requirements

Must Have

- Create incident report
- Edit incident report
- Attach photos
- Auto-extract data with AI
- Categorize incident
- Mark severity
- Track status
- Export PDF
- Search and filter reports
- Show audit history

Should Have

- Dashboard charts
- AI confidence display
- Human correction tracking
- Template preview
- Commenting / notes
- Role-based permissions

Nice to Have

- Repeated incident detection
- Location heatmaps
- AI performance metrics
- Preventive action tracking
- CSV / Excel export

13. Non-Functional Requirements

- Responsive UI
- Fast search and filtering
- Secure file storage
- Role-based access control
- Versioned PDFs

- Audit logs
- Easy template changes
- Support for future claim workflows
- Clear distinction between AI suggestions and human approval

14. Suggested Tech Expectations

Frontend:

- Next.js
- TypeScript
- Tailwind CSS

Backend:

- FastAPI or similar
- REST API
- Auth support
- File storage support

Database:

- PostgreSQL

Storage:

- Object storage for photos and PDFs

AI:

- Vision + structured extraction
- Summary drafting
- Damage / category tagging

15. Acceptance Criteria

The build is successful if:

1. A user can create and edit an incident report.
2. A user can upload multiple photos.
3. AI can suggest incident categories, severity, and tags.
4. A reviewer can correct AI output.
5. The system can show report status clearly.
6. The system can generate a professional PDF.
7. The dashboard can search, filter, and open incidents.
8. Audit logs are recorded for key actions.
9. The design supports future vehicle damage workflows.

16. Important Design Principle

Do not design the dashboard around the PDF.

Design the dashboard around the **structured incident data**.

The PDF should be just one output generated from that structured data.

That makes the system easier to expand later into more advanced damage reporting, claim handling, and AI-assisted review.

17. MVP Scope

For the first version, build:

- Dashboard home
- Incident list
- Incident detail page
- Photo upload
- AI draft fields
- Manual editing
- Status flow
- PDF generation
- Audit log
- Search and filtering

Do not overbuild analytics or advanced claims logic in the first version.

18. Final Note for Builder

The system should feel like a professional incident management tool, not just a form generator.

It must help the team:

- capture incidents quickly,
- review them carefully,
- track them clearly,
- and generate a clean final report.